

GL ReplicatorAgent Sysrisk Project: Model Specification

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Paper: Korinek & Suh, *Scenarios for the Transition to AGI*: <https://arxiv.org/abs/2403.12107>

Model: Scarcity of labor [Equations – Section 2.3, p. 10; Figure/output – Section 2.5, p. 16]

The *Scenarios for the Transition to AGI* paper models how total economic output and wages would change under different scenarios of AGI.

Background

For our paper replication, we are looking at a simpler model that the authors use to build up to their main result, and this model is **not** about the AGI scenarios themselves. As more tasks become automated, output can be produced using either capital or labor for those tasks, while non-automated tasks still require labor.

Output is fully split between:

- payments to labor (the wage bill wL)
- payments to capital (capital income RK): $Y = wL + RK$.

You can see this relationship in Figure 6 below (what we are reproducing):

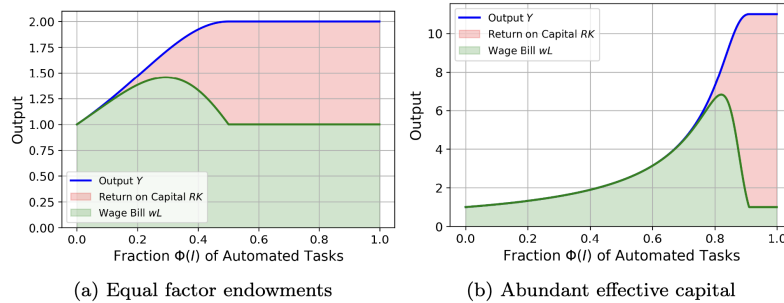


Figure 6: Static equilibria under rising automation

Our Model

The model we are replicating is “scarcity of labor”. This is about how output (Y) and the wage bill (wL) change with the fraction of tasks that can be automated.

This model answers the question of: for fixed capital K and labor L , how do output and factor incomes change as the share of automatable tasks, $\Phi(I)$, rises from 0 to 1?

Equations

Define:

- Y – total output
- A – total factor productivity
- K – capital stock
- $\Phi(I)$ – percentage of total tasks that are less complex than complexity I
- σ – elasticity of substitution
- L – labour endowment
- w – wage; R – return to capital

There is a threshold $\Phi(\hat{I})$, defined by

$$\Phi(\hat{I}) = \frac{\frac{K}{L}}{1 + \frac{K}{L}} \quad (1)$$

such that the economy is in **Region 1** if $K/L > \Phi(I)/(1 - \Phi(I))$, and **Region 2** otherwise. The equations for Y , w , R differ between the two, so the region must be checked first:

$$Y = \begin{cases} A \left[K^{\frac{\sigma-1}{\sigma}} \Phi(I)^{\frac{1}{\sigma}} + L^{\frac{\sigma-1}{\sigma}} (1 - \Phi(I))^{\frac{1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} & \text{Region 1} \\ A(K + L) & \text{Region 2} \end{cases} \quad (2)$$

$$w = \begin{cases} A^{\frac{\sigma-1}{\sigma}} Y^{\frac{1}{\sigma}} L^{-\frac{1}{\sigma}} (1 - \Phi(I))^{\frac{1}{\sigma}} & \text{Region 1} \\ A & \text{Region 2} \end{cases} \quad (3)$$

$$R = \begin{cases} A^{\frac{\sigma-1}{\sigma}} Y^{\frac{1}{\sigma}} K^{-\frac{1}{\sigma}} \Phi(I)^{\frac{1}{\sigma}} & \text{Region 1} \\ A & \text{Region 2} \end{cases} \quad (4)$$

Parameter Table

Symbol	Name	Meaning
K	Capital	units of capital
L	Labour	units of labour
σ	Elasticity of substitution	how easy it is to replace capital (K) with labour (L) (e.g. $\sigma < 1$ means hard to substitute)
A	Total factor productivity	scale parameter (see note)

Parameter	Left figure	Right figure
K (unit of capital)	1	10
L (unit of labour)	1	1
σ (elasticity of substitution)	0.5	0.2
A	0.5	0.2

Looking at Figure 6, we infer that the authors set $A = 1$ in both cases, so we can just set $A = 1$ too.

Input Data

No input data.

Checking Outputs

Basically, we need to calculate Y (and the wage bill wL , capital income RK), and plot how they change with $\Phi(I)$ [see Figure 6, p. 16]. But we should compare values and not just eyeball the graphs:

We need to calculate Y , wL , RK as functions of $\Phi(I)$ [see Figure 6, p. 16]. Checking happens in two stages: first, does *our* implementation reproduce the paper; second, does the *agent's* implementation match *ours*.

Checking the Authors', then Agent's Implementations

First check that our implementation matches the authors', then check the agent's with ours.

- **Threshold:** does the threshold Φ match between the two implementations being compared? If not, stop and fix this first.
- **Spot-check values:** pick a handful of Φ points (e.g. 0, 0.25, 0.5, 0.75, one just above and one just below the threshold, and 0.95) and compare Y , wL , RK at each. They should agree to several decimal places.
- **Shape:** plot Y , wL , RK against Φ . Wages should rise then fall as Φ increases; capital income should do the opposite. In the right figure, the wage bill should drop by roughly 85% shortly before the threshold, matching the paper's description.

If these all look right, treat our implementation as trustworthy for the paper, and use it as the reference for checking the agent.

Next, for the agent specifically:

- **New parameters:** Do the same as above, but use new parameters that we make up ourselves (not in the paper) to confirm the agent actually is generalising from the paper.

Execution Notes

To be updated after meeting with Juan and Shruti tomorrow.

Extra: Some corner values (not exhaustive list)

Expected baseline output values under the baseline parameters ($A = 1$). These are just the obvious ones, not the ones we should use to spot-check:

Quantity	Left figure	Right figure
$\hat{\Phi}$	0.5	$10/11 \approx 0.909$
Y at $\Phi = 0$	1	1
w at $\Phi = 0$	1	1
R at $\Phi = 0$	0	0
Y at $\Phi = \hat{\Phi}$	2	11
$w = R$ at $\Phi = \hat{\Phi}$	1	1

Paper also notes (right figure): wage bill declines by about 85% starting around $\Phi \approx 0.83$.